AUDIT REPORT

llm-cost-router: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/llm-cost-router
Numbers taken from commit	`2ec9b2fce927432c4f8445ad43a97a3a20d56942` (outputs last changed in `2ec9b2f (2026-10-14 12:38:13 +0400)`)
Video	`docs/demo/demo.mp4` · 84.0 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:15
Automated checks	47 pass · 0 fail · 1 manual (of 48 figures)
Assumptions	11 (A-01 to A-11)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/*, outputs/results.json|md, per_case.csv (replays the response cache, under a second)
.venv/bin/python docs/demo/source/tools/export_routing_log.py docs/demo/source/assets/routing_log.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/routing_log.json outputs/results.json docs/demo/source/compositions --rows ext-023,cla-021,sum-026,rea-040,rea-021,cla-035
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/llm-cost-router-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/llm-cost-router-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem', 'Why it is hard'	README lines 12-35	-	general knowledge	✓ verified
F-03	16.4M LLM requests a year / 16.4M / yr / 16.4M req	02, 05, 08	outputs/results.json	annual_requests (sum over 4 tasks = 16,394,979)	sum / 1e6, 1 decimal	derived	✓ verified
F-04	4 tasks	02, 05	outputs/results.json	policy (one entry per task type)	count	direct	✓ verified
F-05	3 self-hosted models / 3 models	02, 05, 10	outputs/results.json	model_seconds (one entry per model)	count	direct	✓ verified
F-06	200 cases, known answers	02, 05, 10	data/eval_suite.jsonl	line count; every case has an 'answer'	count	direct	✓ verified
F-07	3B $0.06 · 7B $0.20 per M tokens	02, 05, 08	src/llm_cost_router/config.py:35-37	MODELS usd_per_m_*	-	direct	✓ verified
F-08	wrong small answers ship silently	02	outputs/results.json	splits.test['router, structural check only'].wrong_accepted_small = 27	> 0	derived	✓ verified
F-09	lower bill / mostly small / answers checked	03	outputs/results.json; src/llm_cost_router/router.py:61	cost_per_1k_usd router < all-large; share_answered_small > 0.5; verify() runs on every small answer	-	derived	✓ verified
F-10	≤ 2 calls / request; FALLBACK	03, 05	src/llm_cost_router/router.py:54-66; README 'Stop condition'	route(): one optional small call, one large call	-	direct	✓ verified
F-11	PO-51719 · Summit Auto Parts · LED-3580 · 20 · Sharjah · 2026-05-14; keep the 3B answer	04, 07	demo:assets/routing_log.json (export of the repo's Router)	id = ext-023: first_model, check_passed, escalated, output, input	-	direct	✓ verified
F-12	6 confident wrong answers got through / 6 wrong answers passed above 0.9999	04, 09	outputs/per_case.csv; demo:assets/routing_log.json	arm = router, tuned on dev; task = reasoning; escalated = False; correct = False	count = 6; every confidence > 0.9999	derived	✓ verified
F-13	Worked for orders, tickets and summaries	04, 09	outputs/results.json	splits.test[arm].by_task[task].completion	router ≥ all-large for 3 tasks	derived	✓ verified
F-14	dev 80 · test 120	05	data/eval_suite.jsonl	split	count per split	direct	✓ verified
F-15	3 models × 200	05, 10	outputs/responses/*.jsonl	line counts	200 lines in each of 3 files	direct	✓ verified
F-16	≈ 46 min CPU	05	outputs/results.json	model_seconds (796.1 + 706.2 + 1240.3 s)	sum / 60 = 45.8 min, rounded	derived	✓ verified
F-17	365 days	05	data/traffic_daily.csv	row count	-	direct	✓ verified
F-18	data.py, eval_suite.jsonl, evaluate.py calibrate(), outputs/responses/, data/, prompts.py, router.py Router.route(), Ollama (client.py), verifiers.py verify(), grading.py, summarise(), paired_bootstrap(), outputs/results.json, per_case.csv	05	src/llm_cost_router/*.py; data/; outputs/	file / def	-	direct	✓ verified
F-19	policy table in Router.route()	05	outputs/results.json	policy[task].first, policy[task].threshold	-	direct	✓ verified
F-20	schema / grounded in input / confidence ≥ τ	05	src/llm_cost_router/verifiers.py:59-149; router.py:62	verify_* functions; verdict.confidence >= policy.threshold	-	direct	✓ verified
F-21	3B kept 91% / 91% of requests / 91%	05, 06, 08	outputs/results.json	splits.test['router, tuned on dev'].share_answered_small = 0.9083	x 100, whole number	direct	✓ verified
F-22	p50 3.6 s · p95 20.4 s	05	outputs/results.json	splits.test['router, tuned on dev'].p50_latency_s, p95_latency_s	1 decimal	direct	✓ verified
F-23	vs 5.8 s for all-7B	05	outputs/results.json	splits.test['all-large qwen2.5-coder:7b'].p50_latency_s	1 decimal	direct	✓ verified
F-24	6 arms · 120	05	outputs/results.json	splits.test arms with a completion	count	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-25	make run < 1 s / replays … in under a second	05, 10	outputs/run_log.txt:85	'make run ... 0.469 total'	total < 1 s	direct	✓ verified
F-26	95% CI · 4,000	05	src/llm_cost_router/evaluate.py:136	reps: int = 4000	-	direct	✓ verified
F-27	per task, the cheapest ≥ 7B	05	src/llm_cost_router/evaluate.py:105-122	options below large completion are skipped; key = (cost, -completion, threshold)	-	direct	✓ verified
F-28	120 held-out requests	06, 07	outputs/results.json	splits.test['router, tuned on dev'].n	-	direct	✓ verified
F-29	109 / 109 answered by 3B / 109 of 120 requests	06, 07, 08	outputs/per_case.csv	arm = router, tuned on dev; escalated = False	count	direct	✓ verified
F-30	11 / 11 escalated	06, 07	outputs/per_case.csv	arm = router, tuned on dev; escalated = True	count	direct	✓ verified
F-31	$0.0176	06, 07, 08	outputs/results.json	splits.test['router, tuned on dev'].cost_per_1k_usd = 0.0175998	4 decimals	direct	✓ verified
F-32	$0.0452	06, 07, 08	outputs/results.json	splits.test['all-large qwen2.5-coder:7b'].cost_per_1k_usd = 0.045235	4 decimals	direct	✓ verified
F-33	82.5%	06, 08	outputs/results.json	splits.test['router, tuned on dev'].completion = 0.825	x 100, 1 decimal	direct	✓ verified
F-34	85.0%	06, 08	outputs/results.json	splits.test['all-large qwen2.5-coder:7b'].completion = 0.85	x 100, 1 decimal	direct	✓ verified
F-35	120-cell grid	06	outputs/per_case.csv; demo:compositions/s04-routes.html (GRID block)	arm = router, tuned on dev: task, escalated, correct	cell-by-cell equality	direct	✓ verified
F-36	failed the confidence bar	06	demo:assets/routing_log.json (export of the repo's Router)	escalated rows: check_passed, confidence < threshold	-	derived	✓ verified
F-37	all qwen 3B 75.8% · all llama3.2 80.0% · all 7B 85.0% · $0.0452 · router, checks only 77.5% · router, tuned 82.5% · $0.0176 · perfect verifier 91.7% · $0.0178	06	outputs/results.json; demo:compositions/s04-routes.html (GRID.arms)	splits.test[arm].completion, cost_per_1k_usd	1 decimal %, 4 decimals $	direct	✓ verified
F-38	61% cheaper / About 61% cheaper / −61%	06, 07, 08	outputs/results.json	cost_per_1k_usd: router 0.0175998, all-large 0.045235	(router / all-large − 1) x 100 = −61.1%	derived	✓ verified
F-39	2.5 points less complete	06, 08	outputs/results.json	completion: router 0.825, all-large 0.85	(0.85 − 0.825) x 100 = 2.5	derived	✓ verified
F-40	6 ledger rows	07	demo:assets/routing_log.json; outputs/per_case.csv; demo:compositions/value.html (SHEET block)	per case: first_model, check_passed, confidence, escalated, output, cost_usd	$ per 1k = cost_usd x 1000, 3 decimals	direct	✓ verified
F-41	+ 114 more requests in the ledger	07	outputs/results.json	n = 120 minus the 6 rows shown	120 − 6	derived	✓ verified
F-42	CI [−7.5, +2.5] pts	08	outputs/results.json	splits.test.router_minus_large_ci95 = [-0.075, 0.025]	x 100, 1 decimal	direct	✓ verified
F-43	$247 / all-7B $779	08	outputs/results.json	splits.test[arm].projected_annual_cost_usd: 246.65, 778.78	whole dollars	direct	✓ verified
F-44	93 / 93 · 87 / 87 · 87 / 83 · 63 / 77	09	outputs/results.json	splits.test[arm].by_task[task].completion	x 100, whole numbers	direct	✓ verified
F-45	38 × 168 = 6264 (true 6384)	09	outputs/responses/qwen2.5-coder_3b.jsonl; demo:assets/routing_log.json	case rea-028	38 x 168 = 6384; 6384 + 45 = 6429	direct	✓ verified
F-46	confidence is read at the ANSWER line	09	src/llm_cost_router/verifiers.py:115-127 (answer_confidence)	lowest token probability among the digits after the final ANSWER marker	-	direct	✓ verified
F-47	OPEN SOURCE · MIT · NO GPU, NO API KEY	10	LICENSE; README 'Running it', 'Inference time'	MIT License; 'need no model server or key'; CPU inference	-	direct	✓ verified
F-48	Ollama · Qwen2.5-Coder · Llama 3.2 · Python	10	src/llm_cost_router/client.py; config.py; pyproject.toml	Ollama client; model ids; python	-	direct	✓ verified

2. Automated check run (2026-10-14 14:15)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	16.4M	16.4M	✓ pass
F-04	4	4	✓ pass
F-05	3	3	✓ pass
F-06	200	200	✓ pass
F-07	true	true	✓ pass
F-08	true	true	✓ pass
F-09	true	true	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	6	6	✓ pass
F-13	true	true	✓ pass
F-14	dev 80 · test 120	dev 80 · test 120	✓ pass
F-15	true	true	✓ pass
F-16	46	46	✓ pass
F-17	365	365	✓ pass
F-18	true	true	✓ pass
F-19	true	true	✓ pass
F-20	true	true	✓ pass
F-21	91%	91%	✓ pass
F-22	p50 3.6 s · p95 20.4 s	p50 3.6 s · p95 20.4 s	✓ pass
F-23	5.8 s	5.8 s	✓ pass
F-24	6	6	✓ pass
F-25	true	true	✓ pass
F-26	true	true	✓ pass
F-27	true	true	✓ pass
F-28	120	120	✓ pass
F-29	109	109	✓ pass
F-30	11	11	✓ pass
F-31	$0.0176	$0.0176	✓ pass
F-32	$0.0452	$0.0452	✓ pass
F-33	82.5%	82.5%	✓ pass
F-34	85.0%	85.0%	✓ pass
F-35	true	true	✓ pass
F-36	true	true	✓ pass
F-37	true	true	✓ pass
F-38	-61%	-61%	✓ pass
F-39	2.5	2.5	✓ pass
F-40	true	true	✓ pass
F-41	114	114	✓ pass
F-42	[-7.5, +2.5]	[-7.5, +2.5]	✓ pass
F-43	$247 / $779	$247 / $779	✓ pass
F-44	93 / 93 · 87 / 87 · 87 / 83 · 63 / 77	93 / 93 · 87 / 87 · 87 / 83 · 63 / 77	✓ pass
F-45	true	true	✓ pass
F-46	true	true	✓ pass
F-47	true	true	✓ pass
F-48	true	true	✓ pass

3. Assumptions register

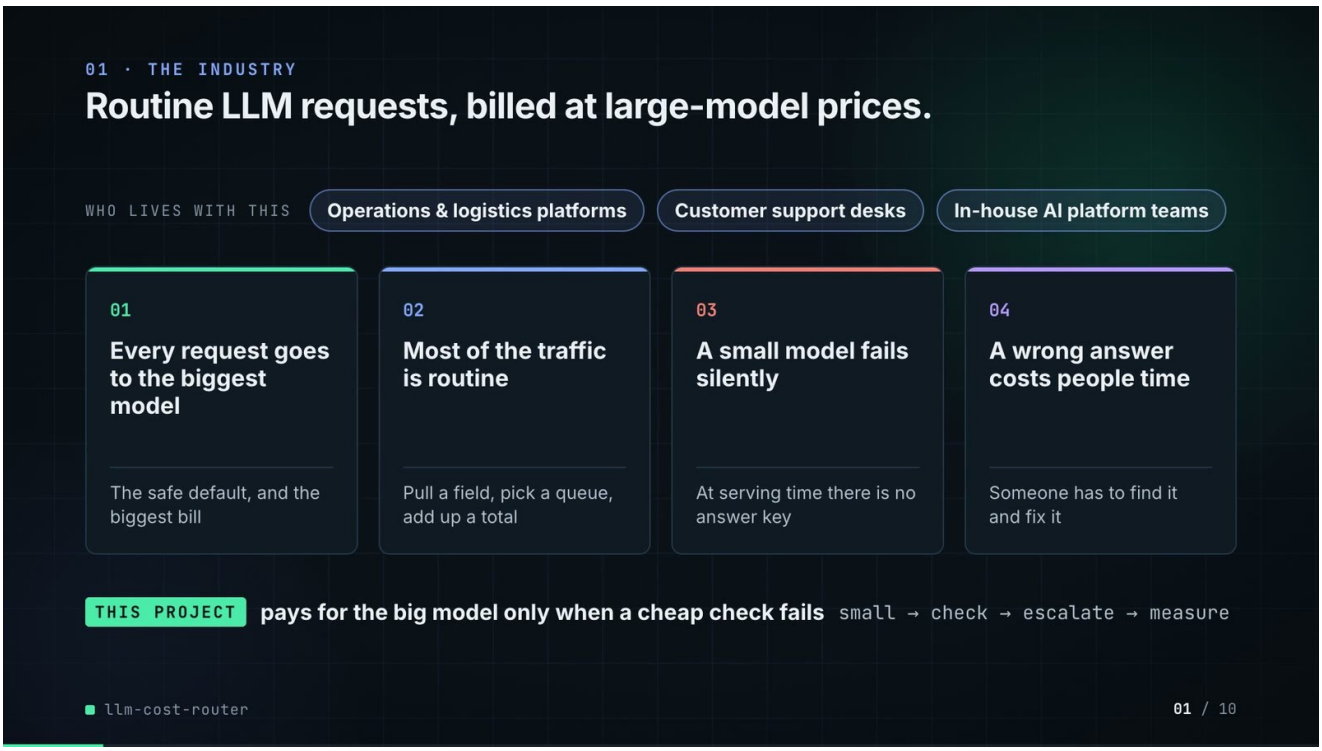
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem: 200 evaluation cases (50 per task type, half easy, half hard) and 365 days of request volume per task type. The production data is proprietary.	src/llm_cost_router/config.py:15 (SEED = 20261013); src/llm_cost_router/data.py:390-470; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Prices are an assumption, not a quote: $0.06 per million tokens for the 3B models and $0.20 for the 7B model. The saving is driven by the 3.3x ratio between tiers.	src/llm_cost_router/config.py:34-37 (MODELS); README 'Known limitations'	F-07, F-09, F-31, F-32, F-37, F-38, F-40, F-43	slides 02, 03, 05, 06, 07, 08
A-03	Latency is the measured CPU inference time per request on an M1 Pro laptop shared with other jobs (model load time excluded); an escalated request waits for the small call and then the large one.	src/llm_cost_router/client.py:79; src/llm_cost_router/router.py:43-46; README lines 53-54 and 'Known limitations'	F-16, F-22, F-23	slides 05
A-04	The routing policy is tuned on the dev split only (20 cases per task type): per task, the cheapest {first small model, confidence threshold} whose dev completion is at least the 7B model's. Every headline number is on the 120 held-out cases.	src/llm_cost_router/evaluate.py:102-134 (calibrate); src/llm_cost_router/data.py:402 (split)	F-12, F-13, F-14, F-19, F-21, F-27, F-28, F-29, F-30, F-31, F-33, F-35, F-36, F-37, F-38, F-39, F-40, F-42, F-44	slides 04, 05, 06, 07, 08, 09
A-05	Ground truth comes from the generator (answers	src/llm_cost_router/	F-08, F-12, F-13, F-33, F-34,	slides 02, 04,

ID	Assumption	Source	Figures affected	Slides affected
	fixed before the text is written); grading.py is the answer key the router never sees. A summary must stay under 40 words and use only numbers from the report.	`grading.py:1-13; src/llm_cost_router/verifiers.py:99-112`	F-35, F-37, F-39, F-42, F-44, F-45	06, 08, 09
A-06	Models run at temperature 0 with seed 7, but multi-threaded CPU inference is not bit-for-bit deterministic; the committed response cache (outputs/responses/*.jsonl) is what makes every number reproducible.	`src/llm_cost_router/client.py:50; README 'Known limitations'`	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-07	Annual volume is the sum of the 365-day traffic log, with logging gaps filled by the mean of the same weekday within 28 days; the projected bill multiplies each task's cost per 1k requests by its volume.	`src/llm_cost_router/evaluate.py:146-166 (annual_volume, projected_annual_cost); data.py:428 (base_daily = 38,000)`	F-03, F-17, F-43	slides 02, 05, 08
A-08	The completion interval is a paired bootstrap over the same 120 cases, 4,000 resamples, seed 0.	`src/llm_cost_router/evaluate.py:136-142`	F-26, F-42	slides 05, 08
A-09	Output caps per task (160, 8, 90, 220 tokens) keep a full collection run under an hour on CPU; a truncated reasoning answer fails its check.	`src/llm_cost_router/config.py:44 (MAX_TOKENS); verifiers.py:134-137`	F-16	slides 05
A-10	Video rounding: percentages to whole numbers or 1 decimal, dollars per 1k to 4 decimals, per-request costs as $ per 1k to 3 decimals, the yearly bill to whole dollars, 16.4M to 1 decimal, 45.8 min shown as '≈ 46 min'.	`this audit (display convention)`	F-03, F-16, F-21, F-22, F-23, F-31, F-32, F-33, F-34, F-37, F-38, F-39, F-40, F-43, F-44	slides 02, 05, 06, 07, 08, 09
A-11	Slide 01 (industry, everyday problems) is general knowledge about running LLMs in operations, used as context. It is not a result of this repo and carries no numbers.	`author's framing; README 'The problem' and 'Why it is hard' support all four problems`	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Routine LLM requests, billed at large-model prices.

WHO LIVES WITH THIS · Operations & logistics platforms · Customer support desks · In-house AI platform teams

01 Every request goes to the biggest model — The safe default, and the biggest bill

02 Most of the traffic is routine — Pull a field, pick a queue, add up a total

03 A small model fails silently — At serving time there is no answer key

04 A wrong answer costs people time — Someone has to find it and fix it

THIS PROJECT pays for the big model only when a cheap check fails · small → check → escalate → measure

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem', 'Why it is hard'` · README lines 12-35	-	✓

Assumptions: A-11

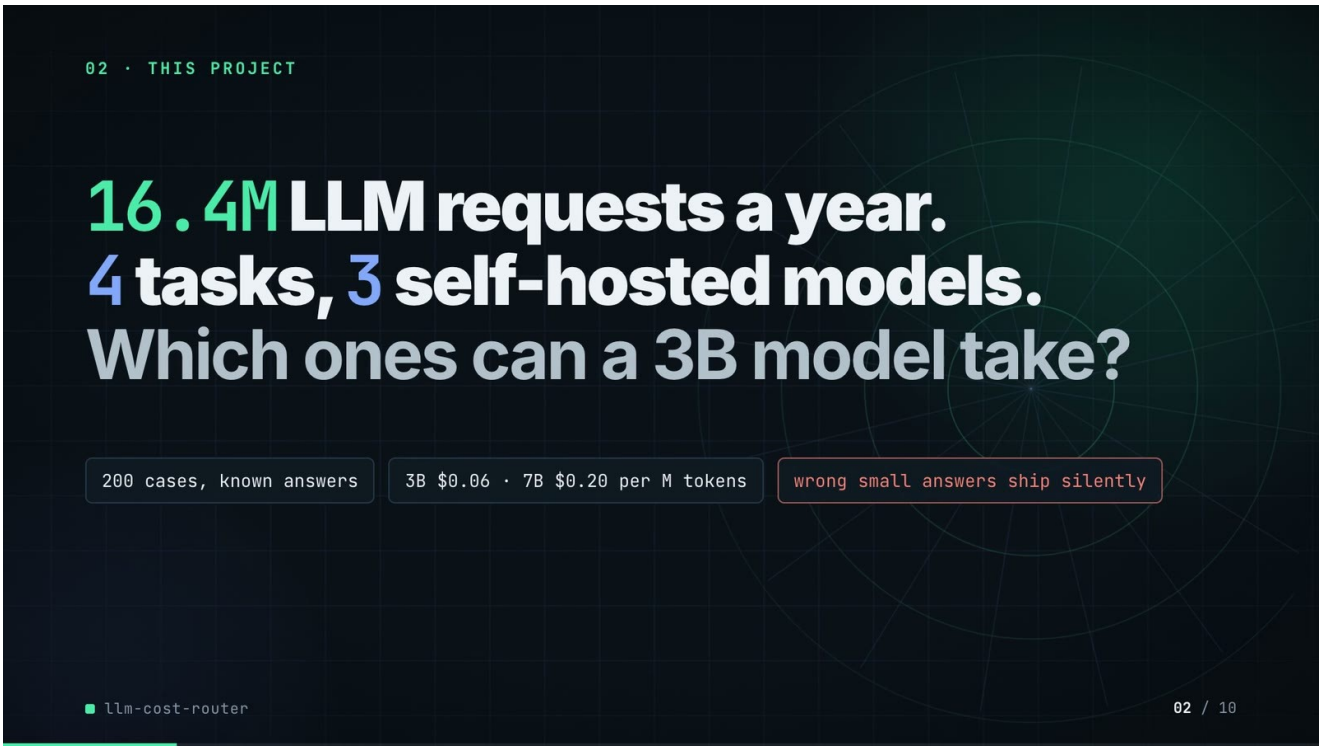
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

16.4M LLM requests a year. 4 tasks, 3 self-hosted models. Which ones can a 3B model take?

200 cases, known answers · 3B $0.06 · 7B $0.20 per M tokens · wrong small answers ship silently

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	16.4M LLM requests a year / 16.4M / yr / 16.4M req	`outputs/results.json` · annual_requests (sum over 4 tasks = 16,394,979)	sum / 1e6, 1 decimal	✓
F-04	4 tasks	`outputs/results.json` · policy (one entry per task type)	count	✓
F-05	3 self-hosted models / 3 models	`outputs/results.json` · model_seconds (one entry per model)	count	✓
F-06	200 cases, known answers	`data/eval_suite.jsonl` · line count; every case has an 'answer'	count	✓
F-07	3B $0.06 · 7B $0.20 per M tokens	`src/llm_cost_router/config.py:35-37` · MODELS usd_per_m_*	-	✓
F-08	wrong small answers ship silently	`outputs/results.json` · splits.test['router, structural check only'].wrong_accepted_small = 27	> 0	✓

Assumptions: A-01, A-02, A-07

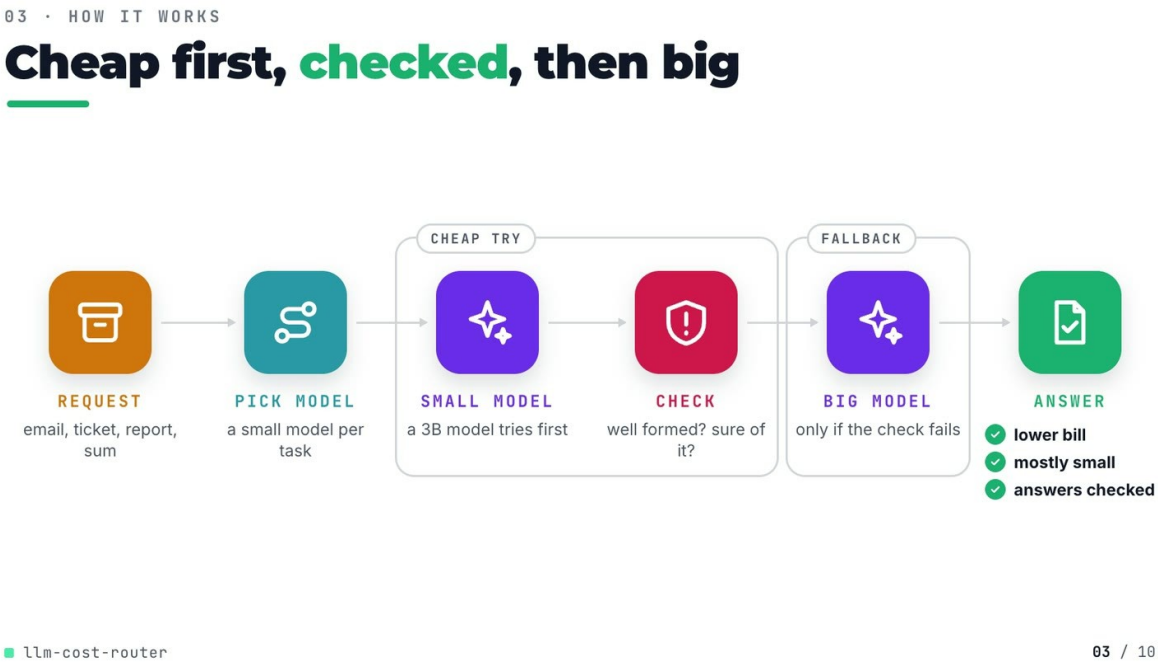
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html.

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups)`



On-screen text (verbatim)

03 · HOW IT WORKS

Cheap first, checked, then big

REQUEST email, ticket, report, sum → PICK MODEL a small model per task → [CHEAP TRY: SMALL MODEL a 3B model tries first → CHECK well formed? sure of it?] → [FALLBACK: BIG MODEL only if the check fails] → ANSWER ✓ lower bill ✓ mostly small ✓ answers checked

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-09	lower bill / mostly small / answers checked	`outputs/results.json`; `src/llm_cost_router/router.py:61` · cost_per_1k_usd router < all-large; share_answered_small > 0.5; verify() runs on every small answer	-	✓
F-10	≤ 2 calls / request; FALLBACK	`src/llm_cost_router/router.py:54-66`; README `'Stop condition'` · route(): one optional small call, one large call	-	✓

Assumptions: A-02, A-04

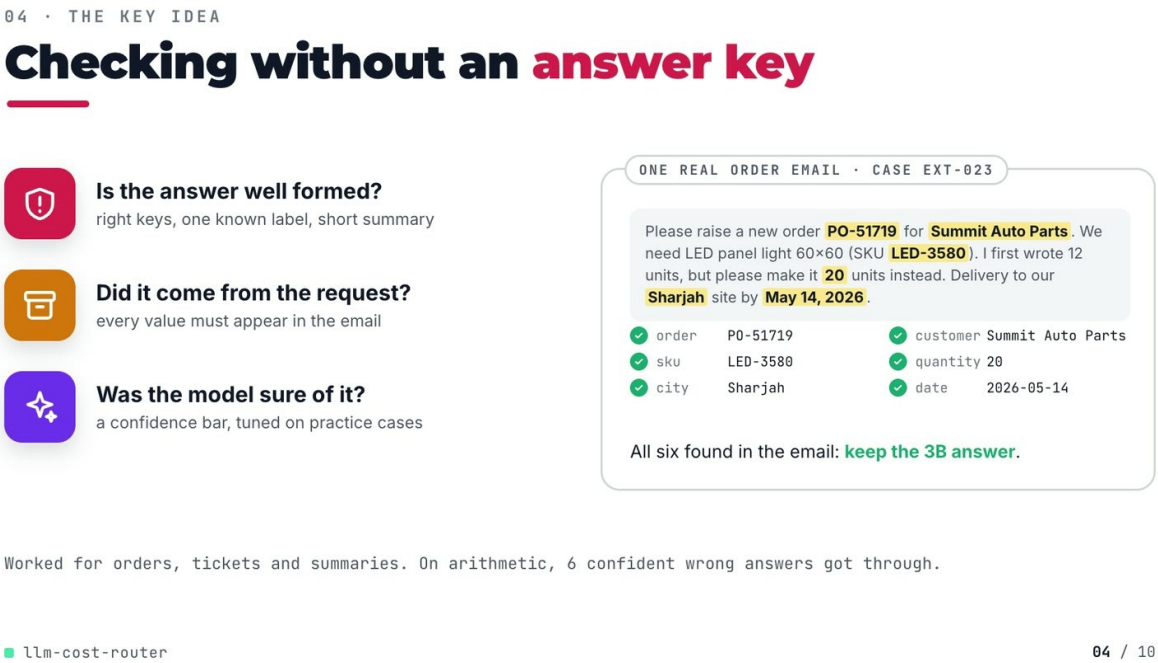
Illustrative, not data:

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth is F-09.

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b)`



On-screen text (verbatim)

04 · THE KEY IDEA

Checking without an answer key

Is the answer well formed? — right keys, one known label, short summary

Did it come from the request? — every value must appear in the email

Was the model sure of it? — a confidence bar, tuned on practice cases

ONE REAL ORDER EMAIL · CASE EXT-023: 'Please raise a new order PO-51719 for Summit Auto Parts. We need LED panel light 60x60 (SKU LED-3580). I first wrote 12 units, but please make it 20 units instead. Delivery to our Sharjah site by May 14, 2026.' → order PO-51719 ✓ customer Summit Auto Parts ✓ sku LED-3580 ✓ quantity 20 ✓ city Sharjah ✓ date 2026-05-14 ✓ · All six found in the email: keep the 3B answer.

Worked for orders, tickets and summaries. On arithmetic, 6 confident wrong answers got through.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-11	PO-51719 · Summit Auto Parts · LED-3580 · 20 · Sharjah · 2026-05-14; keep the 3B answer	`demo:assets/routing_log.json (export of the repo's Router)` · id = ext-023: first_model, check_passed, escalated, output, input	-	✓
F-12	6 confident wrong answers got through / 6 wrong answers passed above 0.9999	`outputs/per_case.csv`; `demo:assets/routing_log.json` · arm = router, tuned on dev; task = reasoning; escalated = False; correct = False	count = 6; every confidence > 0.9999	✓
F-13	Worked for orders, tickets and summaries	`outputs/results.json` · splits.test[arm].by_task[task].completion	router ≥ all-large for 3 tasks	✓

Assumptions: A-04, A-05

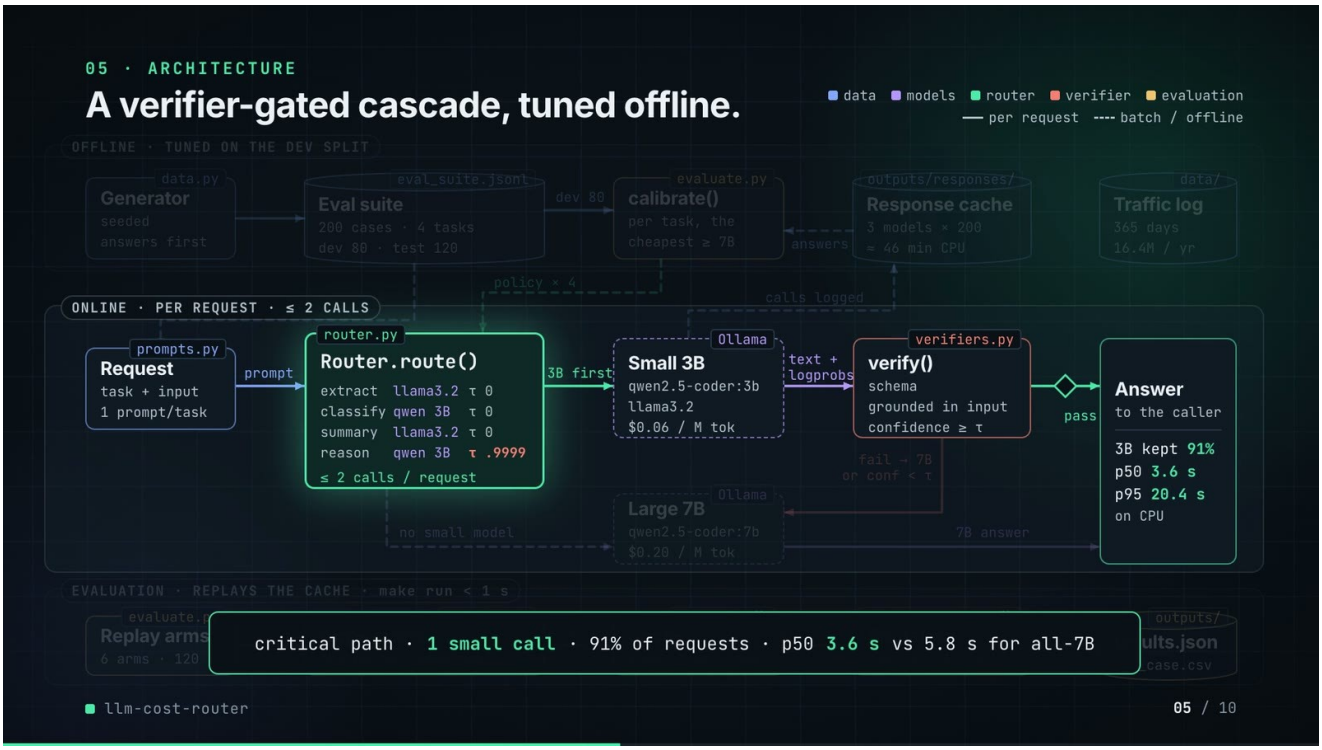
Illustrative, not data:

- The email is an excerpt of case ext-023 (greeting, previous-order line, phone number and sign-off omitted); the six ticks show verify_extraction's grounding rule, which checks every value.

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-12 and F-13.

Slide 05: Architecture

26.0-41.0 s · composition `compositions/s03-architecture.html`



On-screen text (verbatim)

05 · ARCHITECTURE

A verifier-gated cascade, tuned offline.

OFFLINE · TUNED ON THE DEV SPLIT: data.py Generator (seeded, answers first) → eval_suite.jsonl Eval suite (200 cases · 4 tasks, dev 80 · test 120) → dev 80 → evaluate.py calibrate() (per task, the cheapest ≥ 7B) ← answers ← outputs/responses/ Response cache (3 models × 200, ≈ 46 min CPU) · data/ Traffic log (365 days, 16.4M / yr)

ONLINE · PER REQUEST · ≤ 2 CALLS: prompts.py Request (task + input, 1 prompt/task) → prompt → router.py Router.route() [extract llama3.2 τ 0 · classify qwen 3B τ 0 · summary llama3.2 τ 0 · reason qwen 3B τ .9999 · ≤ 2 calls / request] → 3B first → Ollama Small 3B (qwen2.5-coder:3b, llama3.2, $0.06 / M tok) → text + logprobs → verifiers.py verify() (schema, grounded in input, confidence ≥ τ) → pass → Answer (to the caller · 3B kept 91% · p50 3.6 s · p95 20.4 s · on CPU)

fail → 7B or conf < τ → Ollama Large 7B (qwen2.5-coder:7b, $0.20 / M tok) → 7B answer → Answer · no small model → Large 7B · policy × 4 (calibrate → Router) · calls logged (Small 3B → Response cache)

EVALUATION · REPLAYS THE CACHE · make run < 1 s: evaluate.py Replay arms (6 arms · 120) → grading.py Answer key (the router never sees it) → right? → summarise() Score (completion · cost) → paired_bootstrap() Bootstrap (95% CI · 4,000) → outputs/ results.json, per_case.csv

critical path · 1 small call · 91% of requests · p50 3.6 s vs 5.8 s for all-7B

Legend: data · models · router · verifier · evaluation · per request (solid) · batch / offline (dashed)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	16.4M LLM requests a year / 16.4M / yr / 16.4M req	`outputs/results.json` · annual_requests (sum over 4 tasks = 16,394,979)	sum / 1e6, 1 decimal	✓
F-04	4 tasks	`outputs/results.json` · policy (one entry per task type)	count	✓
F-05	3 self-hosted models / 3 models	`outputs/results.json` · model_seconds (one entry per model)	count	✓
F-06	200 cases, known answers	`data/eval_suite.jsonl` · line count; every case has an 'answer'	count	✓
F-07	3B $0.06 · 7B $0.20 per M tokens	`src/llm_cost_router/config.py:35-37` · MODELS usd_per_m_*	-	✓
F-10	≤ 2 calls / request; FALLBACK	`src/llm_cost_router/router.py:54-66; README 'Stop condition'` · route(): one optional small call, one large call	-	✓
F-14	dev 80 · test 120	`data/eval_suite.jsonl` · split	count per split	✓
F-15	3 models × 200	`outputs/responses/*.jsonl` · line counts	200 lines in each of 3 files	✓
F-16	≈ 46 min CPU	`outputs/results.json` · model_seconds (796.1 +	sum / 60 = 45.8 min, rounded	✓

ID	Shown	Source · key	Calculation	Check
		706.2 + 1240.3 s)		
F-17	365 days	`data/traffic_daily.csv` · row count	-	✓
F-18	data.py, eval_suite.jsonl, evaluate.py calibrate(), outputs/responses/, data/, prompts.py, router.py Router.route(), Ollama (client.py), verifiers.py verify(), grading.py, summarise(), paired_bootstrap(), outputs/results.json, per_case.csv	`src/llm_cost_router/*.py; data/; outputs/` · file / def	-	✓
F-19	policy table in Router.route()	`outputs/results.json` · policy[task].first, policy[task].threshold	-	✓
F-20	schema / grounded in input / confidence ≥ τ	`src/llm_cost_router/verifiers.py:59-149`; `router.py:62` · verify_* functions; verdict.confidence >= policy.threshold	-	✓
F-21	3B kept 91% / 91% of requests / 91%	`outputs/results.json` · splits.test['router, tuned on dev'].share_answered_small = 0.9083	x 100, whole number	✓
F-22	p50 3.6 s · p95 20.4 s	`outputs/results.json` · splits.test['router, tuned on dev'].p50_latency_s, p95_latency_s	1 decimal	✓
F-23	vs 5.8 s for all-7B	`outputs/results.json` · splits.test['all-large qwen2.5-coder:7b'].p50_latency_s	1 decimal	✓
F-24	6 arms · 120	`outputs/results.json` · splits.test arms with a completion	count	✓
F-25	make run < 1 s / replays … in under a second	`outputs/run_log.txt:85` · 'make run … 0.469 total'	total < 1 s	✓
F-26	95% CI · 4,000	`src/llm_cost_router/evaluate.py:136` · reps: int = 4000	-	✓
F-27	per task, the cheapest ≥ 7B	`src/llm_cost_router/evaluate.py:105-122` · options below large completion are skipped; key = (cost, -completion, threshold)	-	✓

Assumptions: A-01, A-02, A-03, A-04, A-06, A-08, A-09

Illustrative, not data:

- Packets moving along the wires are a schematic of execution order; 1 escalation in 11 packets approximates the 11 / 120 escalation share, not real traffic.
- 'Ollama' marks the model calls made by client.py; the evaluation replays the same calls from the cache.
- The 'no small model' path exists in router.py (TaskPolicy first=None) but the tuned policy does not use it for any task.
- The critical-path p50 (3.6 s) is the router's median over all held-out requests, including escalated ones.

To change this slide: Node boxes (#n-*) and the edge table E in compositions/s03-architecture.html.

Slide 06: The system running

40.6-53.0 s · composition `compositions/s04-routes.html (GRID block injected by tools/inject_sheet.py)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

Phase 1: The tuned router on 120 held-out requests. · kept the 3B answer · escalated to 7B · ✕ wrong, graded afterwards against the key · rows extraction / classification / summarization / reasoning (30 cells each)

109 answered by a 3B model (of 120 requests) · 11 escalated to the 7B (failed the confidence bar) · $0.0176 per 1k requests (all-7B: $0.0452) · 82.5% graded correct (all-7B: 85.0%)

Phase 2: Six ways to serve the same 120 requests. Completion % vs cost per 1k requests: all qwen 3B 75.8% · all llama3.2 80.0% · router, checks only 77.5% · router, tuned 82.5% · $0.0176 · perfect verifier 91.7% · $0.0178 · all 7B 85.0% · $0.0452; dashed line at the all-7B completion.

Router vs all-7B: 61% cheaper, 2.5 points less complete. The perfect verifier shows the headroom.

Figures on this slide

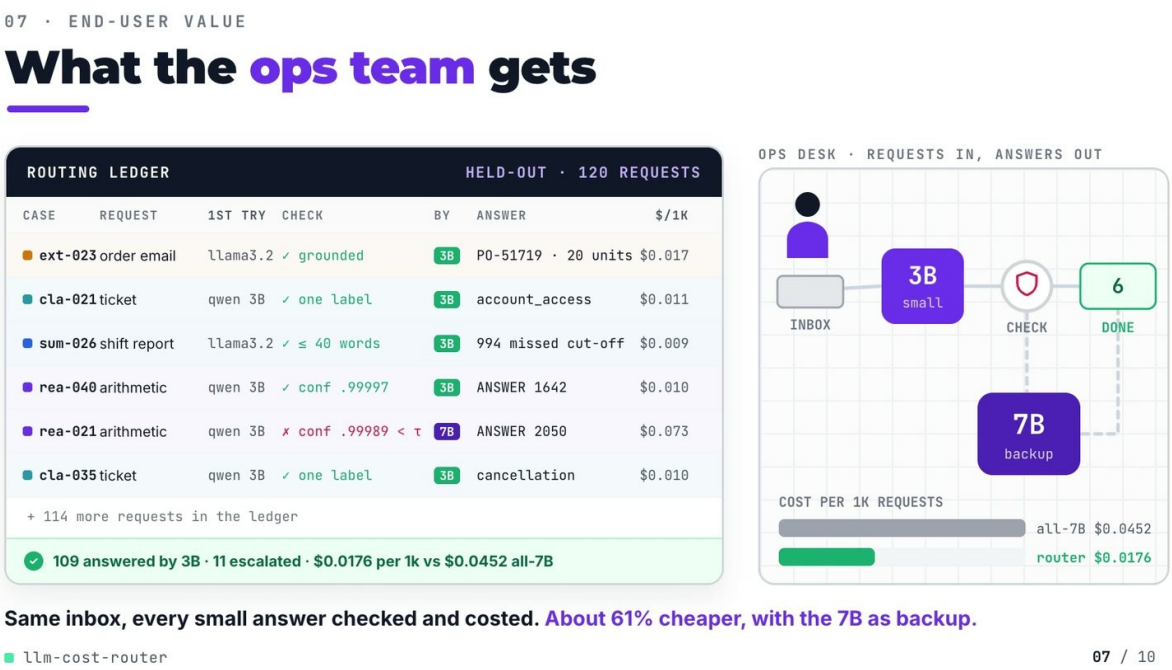
ID	Shown	Source · key	Calculation	Check
F-21	3B kept 91% / 91% of requests / 91%	outputs/results.json · splits.test['router, tuned on dev'].share_answered_small = 0.9083	x 100, whole number	✓
F-28	120 held-out requests	outputs/results.json · splits.test['router, tuned on dev'].n	-	✓
F-29	109 / 109 answered by 3B / 109 of 120 requests	outputs/per_case.csv · arm = router, tuned on dev; escalated = False	count	✓
F-30	11 / 11 escalated	outputs/per_case.csv · arm = router, tuned on dev; escalated = True	count	✓
F-31	$0.0176	outputs/results.json · splits.test['router, tuned on dev'].cost_per_1k_usd = 0.0175998	4 decimals	✓
F-32	$0.0452	outputs/results.json · splits.test['all-large qwen2.5-coder:7b'].cost_per_1k_usd = 0.045235	4 decimals	✓
F-33	82.5%	outputs/results.json · splits.test['router, tuned on dev'].completion = 0.825	x 100, 1 decimal	✓
F-34	85.0%	outputs/results.json · splits.test['all-large qwen2.5-coder:7b'].completion = 0.85	x 100, 1 decimal	✓
F-35	120-cell grid	outputs/per_case.csv; demo:compositions/s04-routes.html (GRID block) · arm = router, tuned on dev: task, escalated, correct	cell-by-cell equality	✓
F-36	failed the confidence bar	demo:assets/routing_log.json (export of the repo's Router) · escalated rows: check_passed, confidence < threshold	-	✓
F-37	all qwen 3B 75.8% · all llama3.2 80.0% · all 7B 85.0% · $0.0452 · router, checks only 77.5% · router, tuned 82.5% · $0.0176 · perfect verifier 91.7% · $0.0178	outputs/results.json; demo:compositions/s04-routes.html (GRID.arms) · splits.test[arm].completion, cost_per_1k_usd	1 decimal %, 4 decimals $	✓
F-38	61% cheaper / About 61% cheaper / −61%	outputs/results.json · cost_per_1k_usd: router 0.0175998, all-large 0.045235	(router / all-large − 1) x 100 = −61.1%	✓
F-39	2.5 points less complete	outputs/results.json · completion: router 0.825, all-large 0.85	(0.85 − 0.825) x 100 = 2.5	✓

Assumptions: A-02, A-04, A-05, A-10

To change this slide: Re-run export_routing_log.py and inject_sheet.py; layout constants in compositions/s04-routes.html.

Slide 07: End-user value

52.7-65.7 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the ops team gets

ROUTING LEDGER · HELD-OUT · 120 REQUESTS · CASE / REQUEST / 1ST TRY / CHECK / BY / ANSWER / $/1K

ext-023 order email llama3.2 ✓ grounded 3B PO-51719 · 20 units $0.017

cla-021 ticket qwen 3B ✓ one label 3B account_access $0.011

sum-026 shift report llama3.2 ✓ ≤ 40 words 3B 994 missed cut-off $0.009

rea-040 arithmetic qwen 3B ✓ conf .99997 3B ANSWER 1642 $0.010

rea-021 arithmetic qwen 3B ✗ conf .99989 < τ 7B ANSWER 2050 $0.073

cla-035 ticket qwen 3B ✓ one label 3B cancellation $0.010

+ 114 more requests in the ledger · ✓ 109 answered by 3B · 11 escalated · $0.0176 per 1k vs $0.0452 all-7B

Scene: OPS DESK · REQUESTS IN, ANSWERS OUT — INBOX → 3B small → CHECK → DONE (count 0..6); rea-021 detours through 7B backup · COST PER 1K REQUESTS all-7B $0.0452 / router $0.0176

Same inbox, every small answer checked and costed. About 61% cheaper, with the 7B as backup.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-11	PO-51719 · Summit Auto Parts · LED-3580 · 20 · Sharjah · 2026-05-14; keep the 3B answer	`demo:assets/routing_log.json (export of the repo's Router)` · id = ext-023: first_model, check_passed, escalated, output, input	-	✓
F-28	120 held-out requests	`outputs/results.json` · splits.test['router, tuned on dev'].n	-	✓
F-29	109 / 109 answered by 3B / 109 of 120 requests	`outputs/per_case.csv` · arm = router, tuned on dev; escalated = False	count	✓
F-30	11 / 11 escalated	`outputs/per_case.csv` · arm = router, tuned on dev; escalated = True	count	✓
F-31	$0.0176	`outputs/results.json` · splits.test['router, tuned on dev'].cost_per_1k_usd = 0.0175998	4 decimals	✓
F-32	$0.0452	`outputs/results.json` · splits.test['all-large qwen2.5-coder:7b'].cost_per_1k_usd = 0.045235	4 decimals	✓
F-38	61% cheaper / About 61% cheaper / −61%	`outputs/results.json` · cost_per_1k_usd: router 0.0175998, all-large 0.045235	(router / all-large − 1) x 100 = −61.1%	✓
F-40	6 ledger rows	`demo:assets/routing_log.json`; `outputs/per_case.csv`; `demo:compositions/value.html (SHEET block)` · per case: first_model, check_passed, confidence, escalated, output, cost_usd	$ per 1k = cost_usd x 1000, 3 decimals	✓

ID	Shown	Source · key	Calculation	Check
F-41	+ 114 more requests in the ledger	`outputs/results.json` · n = 120 minus the 6 rows shown	120 − 6	✓

Assumptions: A-02, A-04, A-05, A-10

Illustrative, not data:

- The ledger is a presentation of the router's real per-request log; the repo writes outputs/per_case.csv, not a ledger file.
- The six rows were picked to show each task and one escalation; they are not a random sample (16 of the 109 kept answers were wrong, see slide 09).
- '994 missed cut-off' condenses the summary's reported count; the full summary text is in the routing log.
- The person, inbox, trays and envelopes are a schematic of the routing flow.

To change this slide: Pick rows with inject_sheet.py --rows; scene geometry in CONFIG in compositions/value.html.

Slide 08: Results

65.4-74.2 s · composition `compositions/s05-results.html`



On-screen text (verbatim)

08 · RESULTS · 120 HELD-OUT REQUESTS
61% cheaper, 2.5 points less complete.
router −61% cost per 1k requests ($0.0452 → $0.0176)
evaluation 82.5% completion, held out (all-7B 85.0% · 95% CI)
models 91% answered by 3B (109 of 120 requests)
projection $247 yearly bill (16.4M req · all-7B $779)
Completion gap CI [−7.5, +2.5] pts · prices are an assumption ($0.06 vs $0.20 per M tokens)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	16.4M LLM requests a year / 16.4M / yr / 16.4M req	`outputs/results.json` · annual_requests (sum over 4 tasks = 16,394,979)	sum / 1e6, 1 decimal	✓
F-07	3B $0.06 · 7B $0.20 per M tokens	`src/llm_cost_router/config.py:35-37` · MODELS usd_per_m_*	-	✓
F-21	3B kept 91% / 91% of requests / 91%	`outputs/results.json` · splits.test['router, tuned on dev'].share_answered_small = 0.9083	x 100, whole number	✓
F-29	109 / 109 answered by 3B / 109 of 120 requests	`outputs/per_case.csv` · arm = router, tuned on dev; escalated = False	count	✓
F-31	$0.0176	`outputs/results.json` · splits.test['router, tuned on dev'].cost_per_1k_usd = 0.0175998	4 decimals	✓
F-32	$0.0452	`outputs/results.json` · splits.test['all-large qwen2.5-coder:7b'].cost_per_1k_usd = 0.045235	4 decimals	✓
F-33	82.5%	`outputs/results.json` · splits.test['router, tuned on dev'].completion = 0.825	x 100, 1 decimal	✓
F-34	85.0%	`outputs/results.json` · splits.test['all-large qwen2.5-coder:7b'].completion = 0.85	x 100, 1 decimal	✓
F-38	61% cheaper / About 61% cheaper / −61%	`outputs/results.json` · cost_per_1k_usd: router 0.0175998, all-large 0.045235	(router / all-large − 1) x 100 = −61.1%	✓
F-39	2.5 points less complete	`outputs/results.json` · completion: router 0.825, all-large 0.85	(0.85 − 0.825) x 100 = 2.5	✓
F-42	CI [−7.5, +2.5] pts	`outputs/results.json` · splits.test.router_minus_large_ci95 = [-0.075, 0.025]	x 100, 1 decimal	✓
F-43	$247 / all-7B $779	`outputs/results.json` · splits.test[arm].projected_annual_cost_usd: 246.65, 778.78	whole dollars	✓

Assumptions: A-02, A-04, A-07, A-08, A-10

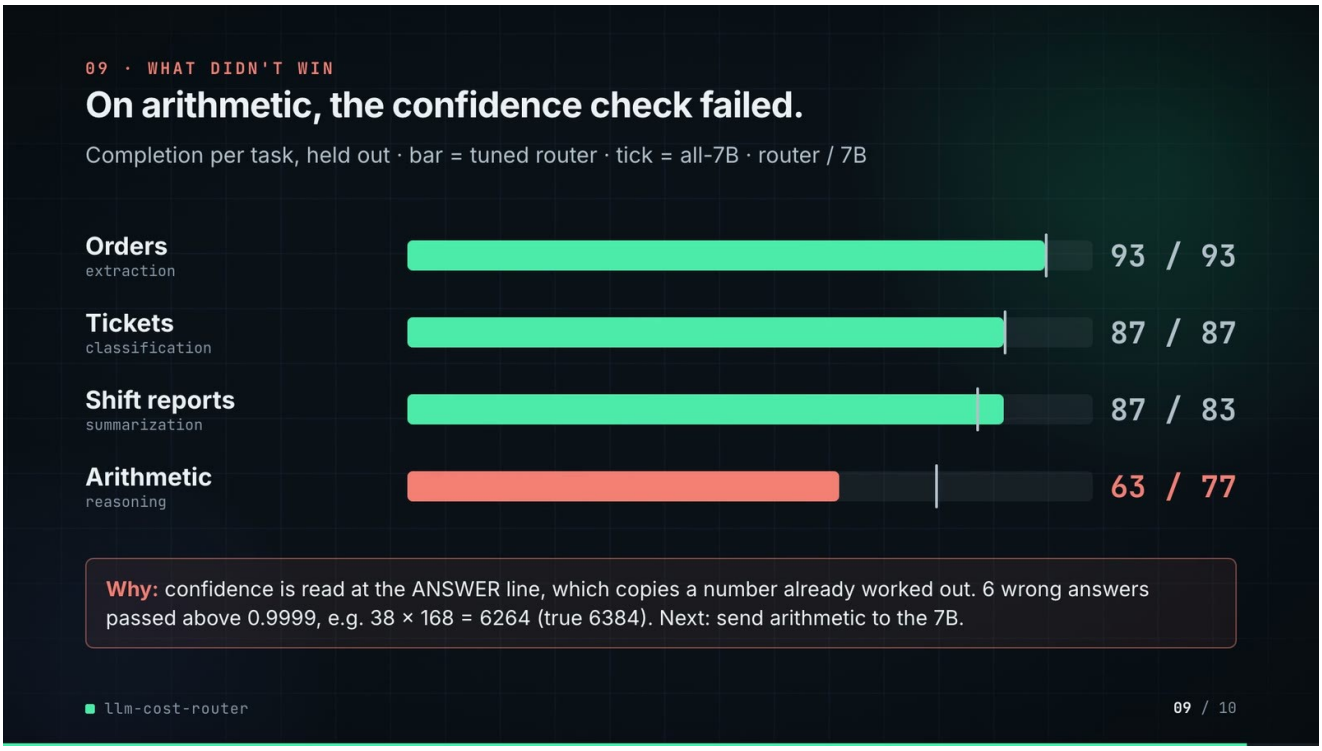
Illustrative, not data:

- Mini bars are ratios to the all-7B baseline (or to 120 requests for the 91% card).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

73.8-80.0 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

On arithmetic, the confidence check failed.

Completion per task, held out · bar = tuned router · tick = all-7B · router / 7B

Orders (extraction) 93 / 93 · Tickets (classification) 87 / 87 · Shift reports (summarization) 87 / 83 · Arithmetic (reasoning) 63 / 77

Why: confidence is read at the ANSWER line, which copies a number already worked out. 6 wrong answers passed above 0.9999, e.g. 38 × 168 = 6264 (true 6384). Next: send arithmetic to the 7B.

Figures on this slide

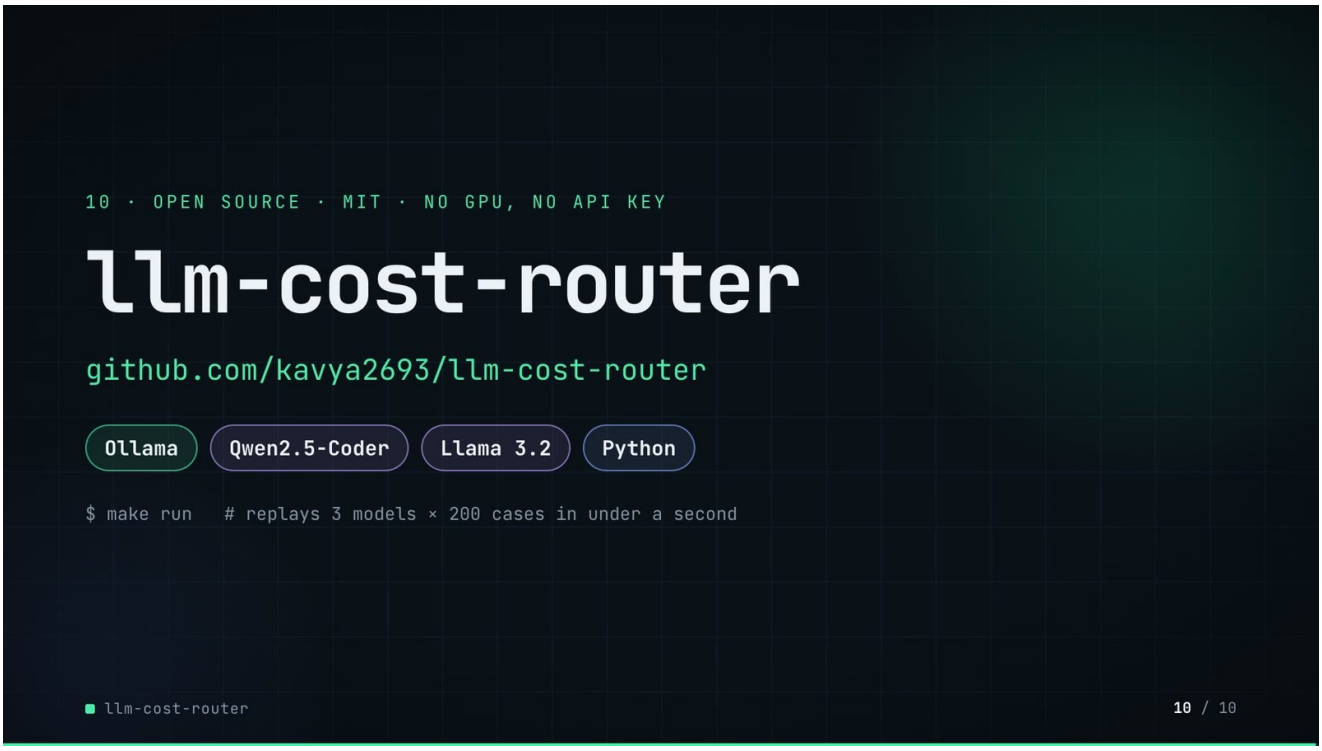
ID	Shown	Source · key	Calculation	Check
F-12	6 confident wrong answers got through / 6 wrong answers passed above 0.9999	`outputs/per_case.csv`; `demo:assets/routing_log.json` · arm = router, tuned on dev; task = reasoning; escalated = False; correct = False	count = 6; every confidence > 0.9999	✓
F-13	Worked for orders, tickets and summaries	`outputs/results.json` · splits.test[arm].by_task[task].completion	router ≥ all-large for 3 tasks	✓
F-44	93 / 93 · 87 / 87 · 87 / 83 · 63 / 77	`outputs/results.json` · splits.test[arm].by_task[task].completion	x 100, whole numbers	✓
F-45	38 × 168 = 6264 (true 6384)	`outputs/responses/qwen2.5-coder_3b.jsonl`; `demo:assets/routing_log.json` · case rea-028	38 x 168 = 6384; 6384 + 45 = 6429	✓
F-46	confidence is read at the ANSWER line	`src/llm_cost_router/verifiers.py:115-127` (`answer_confidence`) · lowest token probability among the digits after the final ANSWER marker	-	✓

Assumptions: A-04, A-05

To change this slide: Bar widths, ticks and values in compositions/s06-twist.html (1000 px = 100%).

Slide 10: End card

79.6-84.0 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · NO GPU, NO API KEY

llm-cost-router

github.com/kavya2693/llm-cost-router

Ollama · Qwen2.5-Coder · Llama 3.2 · Python

$ make run # replays 3 models × 200 cases in under a second

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-05	3 self-hosted models / 3 models	`outputs/results.json` · model_seconds (one entry per model)	count	✓
F-06	200 cases, known answers	`data/eval_suite.jsonl` · line count; every case has an 'answer'	count	✓
F-15	3 models × 200	`outputs/responses/*.jsonl` · line counts	200 lines in each of 3 files	✓
F-25	make run < 1 s / replays … in under a second	`outputs/run_log.txt:85` · 'make run … 0.469 total'	total < 1 s	✓
F-47	OPEN SOURCE · MIT · NO GPU, NO API KEY	`LICENSE; README 'Running it', 'Inference time'` · MIT License; 'need no model server or key'; CPU inference	-	✓
F-48	Ollama · Qwen2.5-Coder · Llama 3.2 · Python	`src/llm_cost_router/client.py; config.py; pyproject.toml` · Ollama client; model ids; python	-	✓

Assumptions: A-06

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem: 200 evaluation cases (50 per task type, half easy, half hard) and 365 days of request volume per task type	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/llm_cost_router/data.py`; `src/llm_cost_router/config.py:15`	make data; make models (~46 min CPU) if prompts or cases change; make run
A-02: Prices are an assumption, not a quote: $0	F-07, F-09, F-31, F-32, F-37, F-38, F-40, F-43	02, 03, 05, 06, 07, 08	`src/llm_cost_router/config.py:34-37`	make run; export_routing_log.py + inject_sheet.py
A-03: Latency is the measured CPU inference time per request on an M1 Pro laptop shared with other jobs (model load time excluded); an escalated request waits for the small call and then the large one	F-16, F-22, F-23	05	`re-collect with make models on the target machine`	make models; make run
A-04: The routing policy is tuned on the dev split only (20 cases per task type): per task, the cheapest {first small model, confidence threshold} whose dev completion is at least the 7B model's	F-12, F-13, F-14, F-19, F-21, F-27, F-28, F-29, F-30, F-31, F-33, F-35, F-36, F-37, F-38, F-39, F-40, F-42, F-44	04, 05, 06, 07, 08, 09	`src/llm_cost_router/evaluate.py:102-134`; `DEV_PER_TYPE` in `src/llm_cost_router/data.py`	make run; export_routing_log.py + inject_sheet.py
A-05: Ground truth comes from the generator (answers fixed before the text is written); grading	F-08, F-12, F-13, F-33, F-34, F-35, F-37, F-39, F-42, F-44, F-45	02, 04, 06, 08, 09	`src/llm_cost_router/grading.py`	make run
A-06: Models run at temperature 0 with seed 7, but multi-threaded CPU inference is not bit-for-bit deterministic; the committed response cache (outputs/responses/*	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`outputs/responses/*.jsonl (via make models)`	make models; make run
A-07: Annual volume is the sum of the 365-day traffic log, with logging gaps filled by the mean of the same weekday within 28 days; the projected bill multiplies each task's cost per 1k requests by its volume	F-03, F-17, F-43	02, 05, 08	`src/llm_cost_router/data.py:428-446`	make data; make run
A-08: The completion interval is a paired bootstrap over the same 120 cases, 4,000 resamples, seed 0	F-26, F-42	05, 08	`src/llm_cost_router/evaluate.py:136`	make run
A-09: Output caps per task (160, 8, 90, 220 tokens) keep a full collection run under an hour on CPU; a truncated reasoning answer fails its check	F-16	05	`src/llm_cost_router/config.py:44`	make models; make run
A-10: Video rounding: percentages to whole numbers or 1 decimal, dollars per 1k to 4 decimals, per-request costs as $ per 1k to 3 decimals, the yearly bill to whole dollars, 16	F-03, F-16, F-21, F-22, F-23, F-31, F-32, F-33, F-34, F-37, F-38, F-39, F-40, F-43, F-44	02, 05, 06, 07, 08, 09	`the composition file named on each slide`	hyperframes render
A-11: Slide 01 (industry, everyday problems) is general knowledge about running LLMs in operations, used as context	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

- The repo's results are on a synthetic suite (A-01); the 16.4M requests a year is the generator's traffic log, not a measured production volume.
- Prices are an assumption (A-02). Every cost figure (61%, $0.0176, $0.0452, $247, $779, the per-row $/1k) scales with the price table.
- The router is 2.5 points less complete than all-7B on the held-out split, with a 95% interval of [−7.5, +2.5]; the slides state the deficit, they do not claim parity.
- 16 of the 109 kept 3B answers were wrong (outputs/results.md 'Wrong answers let through'); the ledger on slide 07 shows six chosen rows, all graded correct.
- Latency is CPU time on a shared laptop (A-03); the router's p95 (20.4 s) is worse than all-7B's (12.6 s) because an escalation pays for two calls. The video shows p95 on slide 05 only.

• routing_log.json is produced by the demo's export script calling the repo's own calibrate() and Router on the committed cache; F-35 and F-40 cross-check it against outputs/per_case.csv.

• Slide 01 statements are general knowledge (A-11).

Appendix: environment

Item	Value
Commit	`2ec9b2fce927432c4f8445ad43a97a3a20d56942`
Python	3.11.7
llm_cost_router	0.1.0
Profile / seeds	suite_and_traffic=20261013, ollama_seed=7, temperature=0, bootstrap_seed=0
Run time	make run replays the committed response cache in 0.47 s (outputs/run_log.txt); refreshing the cache with make models took 45.8 min of CPU inference (13.3 + 11.8 + 20.7 min)